

Review of Distributed Energy Resource and Modern Electrical Grid Optimization Methods

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Abstract—This review synthesizes recent research on convex optimization as a critical tool for managing Distributed Energy Resources and their connections to electrical grid systems. By examining applications in PV-BESS systems, microgrids, and distributed control architectures, this review evaluates how different mathematical formulations, such as Second Order Cone Programs (SOCP) and Linear Programming models (LP), balance computational speed with physical accuracy. Ultimately, this review identifies current limitations in model assumptions and highlights future research needs, with a focus on scalability and integration of diverse renewable energy sources to support a reliable and optimized smart grid.

NOMENCLATURE

Variables

P	Electrical power (kW)
E	Energy (kWh)
t	Discrete timestep/ Time index
Δt	Discrete change in time
G	Prediction Error Tracking
H	Command Penalty
N	Number of time steps
C	Monetary cost
V	Voltage
EM	Emissions
Q	Heat load
S	Economic Savings
U	Control Signal
W	Product between Complex conjugate of Voltage and Voltage
r	Resistance
x	Installed Capacity
y	Admittance
z	Number of units

Subscript

a,b	Node index
g	Technology type
h	Analysis period index
i	participating generator units index
j	Building index
k	Decision step index
l	Trial index
m	Line index
n	Time index
s	Storage index
thr	Threshold Power Rating index

t_{mc}	time of worst constraint violation
xf	Transformer index

Superscript

*	Complex conjugate operator
BESS	Battery Energy Storage System
BIO	Biomass Combined Heat and Power
BUS	Bus
CG	Conventional Generators
CHP	Reserved portion of Combined Heat and Power
e	Electrical Energy
ES	Energy System
f	Forecast
fuel	Fuel
f,PV	Photovoltaic forecast
f,PV+BESS	Photovoltaic and Battery forecast
grid	Electrical Grid
i	Input
l	Load
ME	Maximum Energy
NG	Natural Gas
o	Output
o,PV	Photovoltaic output
o,PV+BESS	Photovoltaic and Battery output
o,grid	Grid output
op	Operations
PV	Photovoltaic System
r	Reduction
T	Transpose Operator
th	Thermal energy
WT	Wind Turbine
FLEX	Characteristic Flexible load

Sets

$\mathcal{A}$	Set of nodes in Network
$\mathcal{B}$	Set of Buses in Network
$\mathcal{D}$	Day index
$\mathcal{G}$	Set of Generators
$\mathcal{M}$	Set of lines in Network
$\mathcal{N}$	Total number of time steps
$\mathcal{H}$	Set of Time periods
$\mathcal{L}$	Set of Trials
$\mathcal{S}$	Set of DES Storage units
$\mathcal{X}$	Set of Transformers

Constants

α	Coefficient of annualized CO_2 emissions reductions (ton/MW)
β	Coefficient of annualized system energy savings (MWh/MW)
γ	Coefficient of annualized fuel savings (MMBtu/MW)
ω	utility coefficient
ϵ	diminishing-return coefficient

I. INTRODUCTION

As environmental concerns and energy consumption pressures grow around the world, the modern electrical grid is experiencing a fundamental transformation [1]. Declining costs for distributed energy resources (DERs) including photovoltaic (PV) technology and battery energy storage systems (BESS) have these technologies [2], grow from rare small scale test projects to a contender for a central feature of new large scale power system developments. DERs are defined as small-scale energy generation and storage technologies located near the point of use. These systems include PV generation devices, BESS, wind turbines, fuel cells, and some combustion turbines and other small thermal generators. DERs can operate independently within microgrids, but can also be interconnected within larger grid networks. As DER penetration continues to rise, system operators and planners face a massive challenge of adjusting to a global fundamental shift in how grid infrastructure and control paradigms are designed [3].

This change also includes that of ensuring reliability of grid operations. The traditional grid setup relies on large synchronous generators with physical inertia allowing for relatively straightforward centralized control to manage frequency and voltage in the wider system. High penetration of DERs has changed this generally predictable system by introducing highly variable inverter-based resources with bidirectional power flows, requiring faster demand response to compensate for the intermittency of renewable-based power generation. This increasingly complex system requires real time management for a greater number of distributed assets than ever seen before. Demand has become apparent for the formulation of new optimization frameworks capable of coordinating DERs efficiently and reliably. This progress towards the smart grid of the future elevates the importance of such frameworks to a central position in both power systems research and energy policy.

A critical dimension occasionally overlooked in this challenge is the role of problem formulation. The methodology for the planning or coordination of DERs, ranging from the choice of objective function, which physical constraints to include, solution methods, and system topology, all play a vital role in the models computational speed and success, as well as the applicability of solutions to the real world. Convex optimization has emerged as a powerful tool in this realm, primarily thanks to the guarantee of globally optimal solutions calculated in small enough time frames to allow for near real time control. Convex relaxations of power flow equations, dispatch models, and DER behaviors allow solvers to reframe

complex nonlinear problems as tractable linear problems. Solution methods including second order cone programming (SOCP) and semi-definite relaxation of optimal power flow are two often used examples of this approach, with each offering key tradeoffs in model fidelity, computational costs, and guarantees of solutions.

This review synthesizes foundational and key recent contributions to the literature on DER optimization with a primary aim of showcasing how model assumptions, objective function design, and constraint structures influence the performance of DER coordination strategies across a range of problem settings. The papers examined span three key focuses; PV and BESS centric distribution systems, microgrid setups, and distributed control architectures. All three domains explore a variety of solution methods, ranging from centralized convex programs solved via interior point methods, distributed consensus based algorithms, to hybrid approaches using machine learning based forecasting and mathematical programming, among others. This review seeks to place these contributions in dialogue with one another and highlight the individual strengths and limitations of these papers, while also identifying gaps in the literature to motivate future research in the field. In doing so, this work seeks to assist in the push towards a future with an optimized DER heavy grid that meets the continually evolving demands of the modern grid.

II. TECHNICAL CONTRIBUTIONS

DERs as a technology have become popular and feasible in the last couple of decades as technologies such as PV and BESS have become more reliable and inexpensive. This means that research on how to optimize these systems is also relatively recent, since earlier deployments were not large enough for planning or operational optimization to be particularly relevant in academia. Early work in distributed generation (DG) planning established the conventional framework for both locational and operational optimization that most papers today still follow. Understanding this history is key to mapping the trajectory of modern power systems away from large centralized generation towards a more diverse distributed one. The basic framework of these early works requires clear links and formulas that connect the feeder topology, load distribution, and the benefits of distributed generation to formulate a plan. The use of sensitivity factors to quantify marginal changes in generation at specific nodes in a DG network laid the foundations for future DER planning methodologies [4].

The modeling of alternating current (AC) power flow has been fundamental to assisting in operational optimization. Replacing nonlinear constraints with semi-definite or branch flow relaxations allows for quickly computed globally optimal solutions for near real time operations [5]. Modern formulations of PV inverter dispatch, voltage regulation, and other operations important to systems with a high penetration of DERs have these convex foundations baked into their functionality. As modern grids deal more with highly complex issues with extreme load conditions, reactive power management, and meshed power networks, relaxation based models may lose their precision. The gaps within this field motivate improved

formulation of convex models to capture the full physics of these DER dominant grid systems.

A. Photovoltaics and Battery Energy Storage System

The first topic of investigation is the use of convex optimization methods with PV and BESS systems. Convex optimization is used to inform planners about the sizing of distributed energy storage (DES), control the dispatch of DES, and minimize operational cost and power loss. This section explores a variety of classes of optimization under different objective functions. The goal is to demonstrate how problem formulation determines the usefulness of the models.

1. Nottrott et al. (LP)

$$\min \{f(P_n^{f,l}, P_n^o)\} = \min \sum_{k=1}^N \left(P_k^{f,l} - P_k^{o,PV+BESS} \right) \Delta t, \quad (1)$$

while $P_k^{f,l} > 0$ and $P_k^{f,l} > P_k^{o,PV}$

Nottrott et al. [6] present an early contribution that provides important historical context for the PV-BESS dispatch optimization problem. The authors formulate a linear programming (LP) approach to determine the optimal battery dispatch of a behind-the-meter PV-BESS system based on day-ahead forecasts, with the objective of peak shaving. The objective function minimizes the difference between the forecast load ($P_k^{f,l}$) and the combined PV-battery output ($P_k^{o,PV+BESS}$) over N (the total number of time steps). The decision variable is the battery dispatch schedule ($P_k^{o,BESS}$), which is a factor of the PV-battery power output ($P_k^{o,PV+BESS} = P_k^{o,PV} + P_k^{o,BESS}$).

This work establishes a foundational LP framework and a benchmark comparison methodology—contrasting forecast-driven optimal dispatch (OPT) with heuristic on-peak/off-peak scheduling (OFFON) and real-time dispatch (RT)—upon which subsequent convex and mixed-integer linear programming (MILP) formulations build. The results indicate that OPT yields a financial advantage over OFFON, primarily due to reduced battery cycling, which extends the expected system lifetime. This effect is particularly pronounced for larger battery capacities and moderate peak-shaving targets, where forecast-driven scheduling more effectively limits unnecessary charge-discharge cycles. In such regimes, the authors report substantial lifetime improvements relative to heuristic scheduling, reinforcing the importance of dispatch strategy in degradation-aware economic performance. This work laid the foundation for later work incorporating multi-objective formulations, stochastic optimization, and more detailed battery models.

2. Li et al. (SOCP)

$$\min f_1 = \min \sum_{n=1}^{\mathcal{H}} \left(\sum_{b=1}^{\mathcal{B}} c_{b,n} P_{b,n}^o + c_{1,n} P_n^{Grid} \right) \quad (2)$$

$$\min f_3 = \min \sum_{n=1}^{\mathcal{H}} \left(\sum_{m=1}^{\mathcal{M}} r_m l_{m,n} + \sum_{xf=1}^{\mathcal{X}} k_{xf} v_{xf,n} + \sum_{s=1}^{\mathcal{S}} P_{s,n}^{DEStloss} \right) \quad (3)$$

$$\min f_4 = \min \sum_n^{\mathcal{H}} \sum_b^{\mathcal{B}} u_{b,n} \quad (4)$$

$$s.t. \quad \left\| \begin{array}{c} 2\sqrt{c_{2b}} P_{b,n}^{BUS} \\ 1 + c_{1b} P_{b,n}^{BUS} + c_{1,n} P_n^{Grid} - u_{b,n} + c_{0b} \end{array} \right\|_2 \leq 1 - c_{1b} P_{b,n}^{GEN} - c_{1,n} P_{Grid,n} + u_{b,n} - c_{0b} \quad (5)$$

Following the focus on foundational dispatch strategies established by Nottrott et al., Li et al. [7] advance the field by addressing the operational and planning challenges of high-penetration PV resources within radial distribution systems. The authors transition from linear programming to a second-order cone programming (SOCP) relaxation, transforming the non-convex optimal power flow problem into a convex framework to determine the optimal allocation and operation of distributed energy storage (DES).

Three objective functions are initially proposed. The first objective, f_1 , minimizes the total operational cost of supplying electricity by combining local generation costs and electricity purchased from the upstream grid. In this formulation, $P_{b,n}$ represents the electrical power produced by at bus b at time period n , while P_n^{Grid} represents the power imported from the electrical grid. The objective is evaluated over all time periods $n \in \mathcal{H}$, making it a standard economic dispatch formulation that seeks the least-cost operating strategy while satisfying network constraints.

The third objective, f_3 , minimizes total system losses. This objective accounts for feeder losses caused by line resistance, transformer losses associated with maintaining voltage levels, and energy losses incurred during DES charging and discharging. Rather than directly minimizing operating cost, f_3 seeks to improve overall network efficiency by reducing energy dissipation throughout the distribution system. This objective is particularly useful in high-renewable systems where reducing losses can improve both system performance and effective utilization of photovoltaic generation.

The objective function, f_4 , is the relaxed cost minimization SOCP function. $u_{b,n}$ is an auxiliary variable that bounds the original quadratic cost minimization for bus $b \in \mathcal{B}$ and time period $n \in \mathcal{H}$, with $c_{1,b}$ and $c_{0,b}$ representing the cost coefficients. By introducing this auxiliary variable and an associated second-order cone constraint, the original quadratic objective can be transformed into an equivalent convex form. Thus, the problem becomes easier to solve while preserving the economic interpretation of the original cost function. The bus generation $P_{b,n}$ and grid power P_n^{Grid} remain constrained by network power balance requirements.

A distinguishing feature of this study is its dual-resolution approach; by operating at two timescales, hourly resolution

Author	Nottrott et al.	Li et al.	Osorio et al.
Objective	Consumer electricity cost	Find optimal DES allocation and operation	Distribution network power loss
Decision Variables	Battery power output	DES size, DES location, DES dispatch	DER location, DER capacity, BESS charge/discharge schedule
Optimization Type	Linear Programming (LP)	Second-order cone programming (SOCP)	Second-order cone programming (SOCP)
Constraints	Power output balance, energy conservation, BESS power limits, BESS charge/discharge limits	PV availability, voltage/line limits, SoC dynamics	Power balance, SoC, ramping, solar resources, inverter capacity
Applied Problem	Behind-the-meter system	Southwestern US	San Andres distribution network
Limitations	No Uncertainty Estimate	Limited Variety of Generator Mix	Relaxation Exactness Unverified

TABLE I
COMPARISON OF OPTIMIZATION APPROACHES FOR ENERGY SYSTEM PLANNING IN PV+BESS SYSTEMS

for sizing and minute-scale resolution for optimal dispatch, the paper addresses the temporal mismatch between planning and operational optimization that many single timescale methods fail to capture. The objective function is formulated to minimize cost and system impacts, such as voltage deviations and power losses, while ensuring the exactness of the SOCP relaxation for radial networks. This work demonstrates that the convex formulation can effectively manage the fast-acting intermittency of PV resources while providing the global optimality required for long-term DES planning. Ultimately, the results underscore the efficacy of using multiple timescales for convex optimization to mitigate the detrimental effects of large-scale distributed PV, providing a computationally efficient alternative to directly solving the original NP-hard non-convex problem.

3. Osorio et al. (SOCP)

$$\min \text{Re} \left(\sum_h \sum_a \sum_b (\|V_{a,h}\|^2 - W_{a,b,h}) y_{a,b}^* \right) \quad (6)$$

Building on the convex optimization foundations discussed by Li et al., Osorio et al. [8] further investigate the utility of SOCPs for optimal power flow (OPF) and DER placement. The authors convert a non-convex bus injection model into a convex SOCP relaxation with the objective of minimizing power losses in distribution networks. Within the objective function, $\|V_{a,h}\|^2$ represents the squared voltage magnitude at node a . $W_{a,b,h}$ is the product of neighboring voltages at node a and b . This quantity is multiplied by $y_{a,b}^*$, the complex admittance of the line connecting nodes a and b . This objective is evaluated over all hours h in the study period $\mathcal{L}$, all nodes a in $\mathcal{A}$, and all connected nodes b in $\mathcal{B}$. This formulation is extended into a multi-period framework,

allowing for the integration of time-varying demand and solar irradiance profiles.

A key technical finding of this work is the quantifiable impact of optimal DER deployment, with results demonstrating that optimal sizing and placement of PV and BESS can reduce network power losses by up to 75%. To validate the robustness of their SOCP approach, the authors conduct a comparative analysis on three different software platforms across four distinct operational scenarios: static power flow, dispatchable DER sizing, PV-integrated profiling, and a comprehensive joint optimization of PV-BESS sizing and operation. This study reinforces the scalability of convex relaxations for loss-minimization objectives and underscores the critical role of spatial allocation in improving the operational efficiency of distribution networks with high DER penetration.

Across these three studies, convex optimization consistently is shown to be effective for PV-BESS dispatch and planning problems, but the choice of formulation carries important tradeoffs. Nottrott et al. establish that even a simple LP framework can significantly outperform heuristic scheduling, while Li et al. demonstrate that transitioning to SOCP unlocks multi-objective and multi-timescale capabilities essential for high-penetration PV systems. Osorio et al. extend this further by showing that SOCP based OPF can simultaneously optimize DER placement and operation for significant loss reductions at the network level. Together, these works showcase a clear progression in problem complexity and solution sophistication, while also highlighting that relaxation exactness and generator mix diversity remain open concerns as these methods are applied to larger and more diverse systems.

B. Microgrids

The next topic of interest is the use of convex optimization in microgrid management. Microgrids are isolated systems that

can operate independently from the main grid for extended periods of time while maintaining an interconnection point for energy exchange. These systems typically have centralized controls to optimize power flows in and out of the microgrid and make electricity dispatch decisions. The development of DERs and the emergence of prosumers introduce complications that require more advanced control systems and modeling to inform investment and planning decisions. As a result, these problems generally include a more dynamic and restrictive sets of constraints due to the diversity of generation, storage, and demand-side resources.

1. Yuan et al. (Co-Optimization)

$$FS = \gamma^{PV} \cdot \sum P_{nom}^{PV} + \gamma^{WT} \cdot \sum P_{nom}^{WT} + \gamma^{Biomass} \cdot P_{nom}^{Biomass} + \gamma^{NG} \cdot \sum P_{thr}^{NG} \quad (7)$$

$$ER = \alpha^{PV} \cdot \sum P_{nom}^{PV} + \alpha^{WT} \cdot \sum P_{nom}^{WT} + \alpha^{NG} \cdot \sum P_{thr}^{NG} + \alpha^{Biomass} \cdot P_{nom}^{Biomass} \quad (8)$$

$$SES = \beta^{PV} \cdot \sum P_{nom}^{PV} + \beta^{WT} \cdot \sum P_{nom}^{WT} + (\beta_e + \beta_{th}) \cdot \left(\sum P_{thr}^{NG} + P_{nom}^{Biomass} \right) \quad (9)$$

The first case considered is the three step co-optimization scheme proposed by Yuan et al. to determine DER sizing in community microgrids. The function problem seeks to maximize three objective functions. The first objective is to maximize fuel savings (FS) as a function of power generation per generation type (P) multiplied by the respective coefficients of annualized fuel savings by generation type (γ), through renewable energy proliferation. The second is maximizing CO_2 emission reduction as a function of generation by type multiplied by the annualized emission reduction by generation type (α). Finally the third objective is to maximize system efficiency as a function of generation by type multiplied by the coefficient of annualized system energy savings (β) to meet state wide energy portfolio mandates [9]. Installed capacities of natural gas combined heat and power (CHP), wind turbine (WT), biomass CHP, and photovoltaic (PV) were used as weights to represent FS as combination of all these factors. This is completed through Karush–Kuhn–Tucker (KKT) conditions and Lagrangian multipliers. The schema next decomposes the net load profile into baseload and high frequency peaker components to assign those demands to CHP and BESS resources to further define necessary DER sizing. The decomposition fo the net load was accomplished by using discrete-time Fourier Transform. Allowing the algorithm to take into account the different timescales at which the various DER technologies respond at. The layout generated is then run through a final optimizer to check against regulations and ensure federal and state mandates are satisfied. This model was validated on a 4.5 MW peak commercial load community in Ohio.

A key feature of this formulation is its sequential optimization structure, which avoids the need for manual weighting in multi-objective formulations. Unlike approaches that embed policy requirements as soft penalties in the objective function, this model explicitly incorporates regulatory constraints. Additionally, the decomposition of the load profile based on temporal characteristics improves the interpretation of demand behavior and enables more effective DER configuration.

2. Cai et al. (Convex MPC)

$$\min \left\{ \sum_{n=1}^H \{ P_n^{grid} \cdot c_n^e + \sum_{j=1}^3 (Q_{j,n}^{ht} \cdot c^{gas}) \} + \sum_{l=1}^L \{ \max(\max_{n \in P_l} (P_n^{grid} - P_{l,k}^{thresh}), 0) \cdot c_l^{DC} \} \right\} \quad (10)$$

The convex optimization based control strategy proposed by Cai et al. focuses on minimizing operational cost by determining optimal schedules for energy flows and building thermal systems [10]. The first sum is the cost calculated as the power purchased from the grid (P_n^{grid}) in the time step (n) multiplied by the cost of electricity in that time step (c_n^e) and the independent heating required ($Q_{j,n}^{ht}$) for the 3 buildings in the study multiplied by the cost of gas (c^{gas}). The second sum is the peak demand cost calculated as the maximum grid demand ($P_n^{grid} - P_{l,k}^{thresh}$) with ($P_{l,k}^{thresh}$) as the peak demand in the grid electricity purchase multiplied by demand-charge coefficient (c^{DC}) imposed by the utility. This convex optimization problem utilizes the model predictive control (MPC) formulation. The cost minimization is subject to battery state of charge, PV generation capacity, and electricity and thermal balance constraints. The charging and discharging of batteries is nonconvex in this formulation, so it was relaxed and included in the efficiency loss terms. The primary purpose was to not allow both discharging and charging simultaneously under optimal conditions. The contribution of this work is in creating a tool that can be used to determine the optimal operation for cost saving and effective sizing for PV and BESS systems. The key advantage of this two stage system is that the MPC allows for optimal scheduling over the prediction horizon rather than instantaneous present time. This allows the proposed strategy to reduce peak demands in electricity with optimal scheduling, flattening the demand curve by staggering cooling times for different buildings in the community. This is a significant contribution because the coordination of energy expenditure between facilities produced significant cost savings when compared to the baseline control strategy used in the case study.

3. Elkazaz et al. (Hybrid Microgrid)

$$C^{OP,MG} = C^{i,MG} - C^{o,MG} + C^{OP,BESS} \quad (11)$$

$$\min_U \frac{1}{2} U^T H_i U + G_i^T U \quad (12)$$

Similar to Cai et al., the work by Elkazaz et al. propose a convex optimization model minimizing daily operating costs for a microgrid, defined as net energy exchange costs with

Author	Yuan et al.	Cai et al.	Elkazaz et al.	Liu et al.
Objective	Multi-objective: fuel savings, emissions, policy compliance	Operational cost minimization	Daily operating cost + degradation cost	Maximize social welfare (utility of consumption minus cost of generation)
Decision Variables	DER capacities, load allocation	Power flows, SoC, thermal control	Dispatch, SoC, grid exchange	Local generation, load consumption, storage dispatch
Optimization Type	KKT-based, multi-stage nonlinear	Convex (MPC)	LP + QP (MPC)	Lagrangian Duality
Constraints	Power balance, policy mandates, DER limits, KKT conditions	Power balance, SoC, PV limits, thermal dynamics	Power balance, SoC, ramping, grid limits, degradation	Power balance, device limits, consensus dynamics
Applied Problem	Ohio microgrid	3-building microgrid	Hybrid PV-wind microgrid	6-bus and 30-bus microgrid systems
Limitations	No Network Constraints or BESS Cycling Constraints	No Network Constraints	Single Model Day Validation Only	No AC Power Flow or BESS degradation protection

TABLE II
COMPARISON OF OPTIMIZATION APPROACHES FOR ENERGY SYSTEM PLANNING IN MICROGRIDS

the grid and BESS operation and degradation costs [11]. The model operates as a two stage controller. In the first stage the LP solved via interior point methods produces 15 minute day-ahead schedules for load, PV, and wind, generating setpoints for the microgrid. The constraints in this stage (power balance, BESS energy dynamics, SoC bounds, and ramping limits) are all affine, which preserves convexity. In the second stage, two real time controllers operating at 1 minute intervals use auto-regression and a QP to minimize tracking error and determine BESS dispatch. The QP objective is composed of the terms from terms H_i , penalizing deviation from the day-ahead schedule, and G_i , penalizing rapidly changing control actions, with U as a vector of BESS control signals to be optimized. BESS degradation is incorporated as a penalty on heavy cycling, expressed as a function of system capital cost and rated cycle life. This rolling horizon structure enables continuous correction of forecasting errors, improving tracking accuracy and operational robustness in real time.

4. Liu et al. (Distributed Fixed-Time)

$$\min \sum_{i=1} f_i(P_i) = \left[\sum_{i=1}^{n_1} C_i^{CG} + \sum_{i=1}^{n_3} C_i^{BESS} - \sum_{i=1}^{n_4} C_i^U \right] \quad (13)$$

$$C_i^U(P_i^{FLEX}) = -\frac{1}{2}\omega_i^{FLEX}(P_i^{FLEX})^2 + \epsilon_i^{FLEX}P_i^{FLEX} \quad (14)$$

Liu et al. proposes a method for solving a Social Welfare Maximization problem, utilizing a microgrid setup containing conventional and renewable generators, BESS, and flexible and constant load devices [12]. In this formulation C_i^{CG} is the cost of conventional generators (i) out of the total number of generators (n_1), C_i^{BESS} is the cost of BESS (i) out of the total number of BESS installed (n_3), and C^U is the utility cost, which can be considered as a representation of customer

benefit over the flexible loads (n_4). The concave cost of utility function is a function of flexible load power (P_i^{FLEX}) for each participating unit (i) weighted by the benefit gained from consuming power (ω_i^{FLEX}) and the diminishing marginal utility coefficients (ϵ_i^{FLEX}) of each participating unit (i). Power balance constraints account for losses in transmission via a linear coefficient while quadratic cost and utility functions tie revenue, cost, utility, and payments together to create a convex program using Lagrangian analysis. KKT generates a market clearing price from the optimization model which all network participants choose to dispatch at. The market clearing price is updated constantly while enforcing supply and demand balances. Local decision making for each device shares dispatch choices and market clearing estimates with nearby devices. An upper bound consensus timescale can be derived from the problem formulation, one dependent on the algorithm and network structure, rather than on any initial conditions. Validation testing was performed on 6 and 30 bus systems examining simple single day scheduling and more complex "plug and play" scenarios. The key difference between this work and the previous cases is that this model functions very similarly to real markets by using a universal clearing price to determine dispatch between many interconnected participants.

In these studies, convex optimization emerges as a unifying framework for both centralized and distributed microgrid management. However, the specific model formulation choices reflect fundamentally different system design considerations and what was important to the respective researchers. The social welfare formulation represents an extension within the field by introducing new constraints rooted in market-based coordination [12]. These works demonstrate the evolution of control systems by expanding the PV BESS problem examined previously into more commercialized distribution architectures with centralized control.

C. Distributed Control Algorithms

As the last case in the previous section alluded to, the distribution of generation capabilities down to the consumer level creates a need for more decentralized control. In parallel, regulatory frameworks such as the Energy Performance of Buildings (EPBD) in the EU [13] have established requirements for sustainable building development. These problems consider the interaction between prosumers, DERs, and the grid, highlighting the need for intermediate control systems at the building level.

1. Lindberg et al. (MILP ZEB)

$$\min \pi = \sum_{i \in I} (C_i^{\text{tot spec}} x_i + C_i^{\text{tot fixed}}) + \sum_{h=1}^H \frac{1}{(1+r)^{(h-1) \cdot N(h)}} \cdot \sum_{n=1}^N \frac{C_h^{\text{tot run}}}{(1+r)^n} [EUR] \quad (15)$$

The work in Lindberg et al. considers the EU definition of a near zero energy building to optimize investment technologies and sizing as well as the operation of the power system within the building [14]. The nomenclature for this problem formulation consists of the discounted specific investment cost including reinvestment for each technology i , $C_i^{\text{tot spec}}$, discounted fixed investment cost including reinvestment for each technology i , $C_i^{\text{tot fixed}}$, and annual operational costs for a representative year in an h period, $C_h^{\text{tot run}}$. This study balances reduction in the energy demand and on-site generation through the lens of minimizing cost. The global cost function encompasses initial investment cost, operational, and replacement costs across the full building life cycle. The model takes hourly loads for heat and electricity, self generation for PVs, building design including the u -values and HVAC specifications, and strictness of ZEB adherence as parameters for the problem. Both continuous and binary variables are simultaneously used throughout the MILP algorithm to optimize component sizing, technology selection and scheduling of operations for each of these technologies. The paper uses MILP formulation to minimize economic cost of investment and operation for buildings under different levels of ZEB criteria. Because this is a non-convex problem however, the optimal solution is not guaranteed. This MILP formulation requires all constraints and objectives to be linear. Due to this constraint, all dynamics of the power flow problem and cost evolution were modeled as linear. As a result, a globally optimal result was approximated with respect to this model. In more recent years, a new formulation was studied which includes nonlinear convex functions.

2. Muñoz-Salcedo et al. (MIDCP)

$$\min_{z,x} \sum_i^G C_i^{\text{inv}} z_i + \sum_n^H \sum_i^G C_i^{\text{op}} x_{i,n} + \sum_i^G C_i^{\text{rep}} z_i + \sum_n^H c^{CO_2} EM_n \quad (16)$$

The novelty in the approach outlined by Muñoz - Salcedo et al. is in adding life cycle analysis to the different energy

sources in order to create a tool that can be used for long term sizing and planning of net zero buildings [15]. The original global cost problem is modified to analyze discrete, continuous and binary variables transforming it into a Mixed Integer Disciplined Convex Problem (MIDCP). The decision variables are the number of sources of each technology to be installed (z) and the selected operation energy produced (x). The first cost associated with this formulation is the initial investment cost (C_i^{inv}) of each selected technology (z_i) in the set of technology types (G). The second cost is the sum of the operational costs of each technology type (C_i^{op}) and the selected operational capacity of each technology (i) in the set (G) for each step (n) in the time period (H). A unique feature of this formulation is that the replacement costs are also added to the global economic cost function (C_i^{rep}) for each selected (z_i). Finally a penalty price on carbon emissions is incurred by multiplying carbon penalty factor (c^{CO_2}) by the emissions (EM_n) over the time horizon. The problem parameters include technological parameters of considered technologies, performance and generation profiles, battery storage, building demand profile, price of carbon emissions and climate conditions. Both formulations involve integer decision variables in the quantity of technology installed which makes the problem non-convex. While this is not a convex formulation, the problem follows disciplined convex rules which allows it to be pseudo-convexified using the solver and has been shown to converge to the global optimum. The algorithm has been shown to achieve global optimal convergence quickly using two case studies in Madrid and Ecuador.

Both Lindberg et al. and Muñoz-Salcedo et al. address the building scale energy planning problem through the lens of life cycle cost minimization, and together they illustrate an important progression in problem formulation. The MILP approach of Lindberg et al. provides a baseline that enforces ZEB constraints through linear approximations, while the MIDCP framework of Muñoz-Salcedo et al. extends this by incorporating nonlinear convex cost terms, additional storage technologies, and emission penalties within a convex model structure. Both works serve as examples of the necessary balance between model quality and solution guarantees inherent to mixed-integer formulations, and neither study addresses scalability beyond the single building level, leaving multi-building or community scale extensions as a clear direction for future work.

III. LIMITATIONS

Convex optimization has become a central tool for coordinating DER dominant electrical setups, but current formulations still have important limitations that affect the applicability of their results. Despite rapid growth in installations globally, DERs still represent a modest share of total grid generation and storage capacity. The recency of many deployments and optimization solutions calls into question how practical those findings will end up being in proposed future projects.

In the works analyzed here, most limitations are a question of problem formulation and assumptions. While making model

Author	Muñoz-Salcedo et al.	Lindberg et al.
Objective	Lifecycle cost minimization	Lifecycle ZEB cost minimization
Decision Variables	Integer tech selection, continuous sizing	Binary tech selection, sizing
Optimization Type	MIDCP	MILP
Constraints	Energy balance, SoC, emissions, integer limits	ZEB constraints, building physics
Applied Problem	Madrid / Ecuador	ZEB design
Limitations	Restrictive Convexity Assumptions	Low Model Resolution

TABLE III
COMPARISON OF OPTIMIZATION FRAMEWORKS FOR DISTRIBUTED CONTROL ALGORITHMS

reductions and relaxations is inherent to the convexification of real life systems, these works are only as useful as the conditions under which the results are feasible and applicable. Works in this collection acknowledge this importance, and outline scenarios in which their relaxations will recover the true optimal solutions [5]. SOCP and SDP relaxations may yield solutions that are feasible for the formulated problem but physically infeasible in real electrical systems if relaxation exactness conditions are not met [16]. Incorrect solutions to convex optimization can also be hard to notice without comparison against solutions to the original nonconvex problem. As DERs become further incorporated into modern grids and face more extreme operating conditions, including highly variable, non-periodic load profiles associated with large data centers or electric vehicle charging infrastructure, the chances of exactness conditions being violated increases.

Another model simplification comes from the cornerstone DER technology that are BESS. While ramping and thermal conditioning is typical to conventional thermal generator modeling, BESS systems were often not provided the same treatment in these works. To ensure long term battery health and reliability, manufacturers recommend strict operation procedures for real life systems [17]. These recommendations maximize lifetime, reduce downtime, and maximize performance metrics over a systems lifetime. While some works introduce penalties into the cost function [11], most papers only limited BESS within feasible state of charge metrics and output wattage [18]. Since real life BESS devices generally operate under their manufacturers' parameters, ensuring model parameters do not invalidate their output before optimization is even performed is imperative. An improperly conditioned model can force heavy cycling commands onto a BESS and potentially introduce safety hazards [9].

As mentioned earlier in this review, DER systems have the potential to introduce fast dynamic devices to the grid system, allowing for high speed demand response and frequency regulation. While some devices have internal technical capacity to make power injection decisions at high speeds and even become grid forming after a fault event, this is not always allowed by regulators or within device parameters. This responsibility then falls to the optimization model to ensure

proper decisions can be made in real time to aid in grid management. This consideration was often inadequately addressed in two distinct ways. The first being that modeling speeds are not always a factor in research goals, with convergence times being allowed to inflate with no limits [11]. Secondly, some quicker models have high speed convergence but lack analysis on how discretization may affect solutions [12].

Physical system testing has become a more common component of DER optimization modeling, with some lab setups including small BESS utilized within hardware-in-the-loop style testbeds [11]. Generally, most other papers validate through numerical solutions using benchmark studies and historical data. While these tests provide useful insight into model efficacy, they can be hindered by copper-plate assumptions. Copper-plate assumptions place all simulated generators and storage devices onto a single node or bus, effectively removing power dynamics like line impedances, voltage dynamics, and reactive power flows [19]. Those metrics are key to operational feasibility, and reductions in this order, while making optimization solving much simpler, force extreme restraints on result applicability.

IV. UNANSWERED QUESTIONS AND EXTENSIONS

A. Foundational Methods

A common theme in the foundational papers were problems that arise from them being so novel for their time. These papers set the bases for much of the literature in this project. Thus, it's unsurprising to find there are several areas where questions were left unanswered. For example, after going over the findings shown in the very first paper in terms of chronological order, Acharya et al. showed some of the first algorithms that considered the optimal size of DG to be just as important as its bus location in order to minimize losses [20]. This paper begged the question of how much more efficient these algorithms could become. Both in terms of speed and computational cost of the procedures to ensure large scalability of the model and in terms of its complexity of the factors or physical properties considered of the system to improve the accuracy of the results. The foundational paper that helped to steer these grid distribution problems into the

realm of convex optimization was the study done by Lavaei & Low [5]. They showed how these NP-hard problems could be effectively solved by relaxing nonconvex problems to an equivalent problem formulation that was convex. Resulting in approximate solutions that were in many cases as good as the exact answer. This naturally leads to a curiosity of how broadly applicable this method of relaxing nonconvex problems is to energy management. What other methods exist that follow a similar framework and how much more accurate or scalable are they. Specifically for this paper, an extension of the field would be to explore how a second order cone programming may be applied to the semi-definite relaxed problem in order to expand usability on a system in orders of magnitude of thousands of buses. When looking at modern literature, a lot of the same themes are still present when taking a broad look at what unanswered questions remain. However, as a result of the accomplishments made over the past decade much of these questions have become more focused and have honed in on specific technical details that need to be further explored.

B. PV and Battery Energy Storage Systems

In the scope of photovoltaic and battery energy storage systems one of the common set of questions remaining after examining the literature is how these methods could integrate with other forms of variable renewable energy (VRE) systems outside of just PV systems into the model. For example, there are several locations in the US where wind energy is greatly accessible as it has quickly become one of the cheapest forms of renewable energy. In the case of Li et al. study they tested the interesting case of applying two time scales to increase the accuracy [7]. It will be interesting to see how these time-scales may have to change to account for the additional renewable energy systems, especially with how unpredictable the time generation of wind energy can be. A potential solution to this problem may be to use machine learning techniques to conduct weather forecasting that would allow researchers to better estimate not only wind speeds but also its pathing. In the case of an extra paper not included in the review, but still relevant in this context, Georgiadis et al. [21] had deep learning algorithms are baked into their methods, forecasting was already being done. But, it was for the case of photovoltaic systems power generation and economic costs of buy/selling electricity. One of the previously mentioned limitations of this study was the lack of data the research team has access to for training their model. However, there are immense amounts of wind data available through various government agencies such as the DOE and NOAA, that could be used to support the implementation of wind forecasting models. Thus, making wind an enticing application for this method. For the common theme of scalability, the work conducted by Osorio & Garcia falls under this category [8]. The model proved to be extremely successful to reduce power loss by $\frac{3}{4}$ in smaller sized systems. But, the stochastic routine algorithms used in the model are known to be sensitive to problem size. It would be interesting to see exactly the drop in efficiency for larger systems, and what possible alternative relaxation methods could be applied to potentially alleviate these issues as the model showed very promising results at smaller scales.

C. Microgrid Energy Management

When analyzing the subsection of literature on microgrids, the concept of expanding models to collaborate with other proven algorithms is ever present. This is particularly true for the method derived by Yuan et al. [9]. The model's use of three successive objective functions proved to be fruitful for maximizing fuel savings. However, the lack of considerations of optimizing power flow begs the question on how easily an SOCP relaxation optimization could be added to the existing method given how well it already handled three objective functions. Another interesting aspect of this paper is how it strongly enforced constraints to model government regulations. This leads to speculation on how this model will hold as new mandates are constantly being added due to the rapid growth of the industry. Another paper that used a variety of different algorithms to optimize various components was literature authored by Elkazaz et al. [11]. However, their MATLAB simulations showed the results of their algorithms for a perfectly ideal case. Thus, the model's robustness is still in question as a real world scenario would have disturbances such as delays, converter dynamics, and seasonal variability in generation and load profiles.

D. Distributed Control Architecture

The final branch examined in this review, distributed control architecture, contains a similar structure optimization algorithm as many of the previous works and also relies on mixed integer programming. The interesting aspect about this section is that work done by Munoz - Salcedo et al. heavily builds on the research conducted by Lindberg et al. [15], [14]. The initial paper by Lindberg showed how thermal storage can be used to optimize the efficiency of a building. However, thermal was the only type of storage that was considered in this study. It raised the question how others could potentially be incorporated. Munoz-Salcedo accomplished integrating lithium-ion battery storage and hydrogen storage in addition to thermal storage by using a more complicated MIDCP algorithm. An unexplored question for both of these studies is how their work can be scaled up to incorporate multiple buildings or perhaps even a small neighborhood. Additionally, with one of the main goals of these two papers being to improve communication between nodes in a decentralized grid, it's a wonder how it would be affected by policies concerning privacy of generation operations? As protection of optimal solution communication is often a topic of great concern in cybersecurity.

V. CONCLUSIONS

This review examined the evolution of convex optimization methods across three levels of energy systems of varying complexity: PV+BESS, microgrids and distributed control frameworks. A range of formulations including MILP, MIDCP, KKT-based methods, MPC, OPF, BFM, and convex relaxations such as LP, QP, and SOCP were discussed and shown to address objectives spanning operational and lifecycle cost, emissions reduction, loss minimization, and social welfare.

A central challenge across a majority of approaches is the inherent non-convexity of realistic energy system models,

which have been addressed through relaxation methods or structure reformulations that trade fidelity for tractability.

Future work in this industry will continue improving convexification techniques and extending models. Emerging applications such as data center loads and electrification of the transportation sector will further expand the scope and complexity of these optimization problems and drive continued advancement in convex optimization methods for energy applications.

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